

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

AI
Practical – VII TASK

Roll No.: T038	Name: Akansha Pritam Ranade
Class: TYIT	Batch: 1
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Q.1

Area (sq.ft.)	Price (Lakhs)
600	32
800	40
1000	49
1200	58
1400	68
1600	79

Find the regression equation.

Predict the price of a 1500 sq.ft. house.

Predict the price of a 2500 sq.ft. house.

```
import numpy as np
```

```
from sklearn.linear_model import LinearRegression
```

```
# Given Data
```

```
X = np.array([600, 800, 1000, 1200, 1400, 1600]).reshape(-1, 1) # Area
```

```
y = np.array([32, 40, 49, 58, 68, 79]) # Price in Lakhs
```

```
# Train the model
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

LinearRegression ⓘ ?		
Parameters		
	<code>fit_intercept</code>	True
	<code>copy_X</code>	True
	<code>tol</code>	1e-06
	<code>n_jobs</code>	None
	<code>positive</code>	False
Fitted attributes		

Fitted attributes		
Name	Type	Value
<code>coef_</code>	ndarray[float64](1,)	[0.05]
<code>intercept_</code>	float64	2.79
<code>n_features_in_</code>	int	1
<code>rank_</code>	int64	np.int64(1)
<code>singular_</code>	ndarray[float64](1,)	[836.66]

```
# Get Equation Parameters
```

```
slope = model.coef_[0]
```

```
intercept = model.intercept_
```

```
print(f"Regression Equation: Price = {slope:.5f} * Area + {intercept:.2f}")
```

```
Regression Equation: Price = 0.04686 * Area + 2.79
```

```
# Predictions
```

```
areas_to_predict = np.array([1500, 2500]).reshape(-1, 1)
```

```
predictions = model.predict(areas_to_predict)
```

```
print(f"Predicted price for 1500 sq.ft.: {predictions[0]:.2f} Lakhs")
```

```
print(f"Predicted price for 2500 sq.ft.: {predictions[1]:.2f} Lakhs")
```

Predicted price for 1500 sq.ft.: 73.08 Lakhs
Predicted price for 2500 sq.ft.: 119.93 Lakhs

Q.2 A company has collected the following employee data:

- Employee ID
- Employee Age
- Years of Experience
- Salary

You need to develop a Linear Regression model to predict an employee's salary. Tasks

1. Identify the independent variable (X) and the dependent variable(Y).

Independent Variable (X): Years of Experience

Dependent Variable (Y): Salary

2. Explain why you selected these variables.

Years of Experience is selected as the independent variable (X) because it is the input feature used to predict the employee's salary. The number of years an employee has worked is expected to influence their salary.

Salary is selected as the dependent variable (Y) because it is the output or target variable that depends on the employee's years of experience.

The Linear Regression model learns the relationship between experience and salary to predict salary values for new employees.

3. Create a Pandas DataFrame using the given data.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
data = {
    'Employee ID': [1, 2, 3, 4, 5, 6, 7],
    'Employee Age': [22, 25, 28, 30, 35, 40, 45],
    'Years of Experience': [1, 2, 4, 5, 8, 10, 12],
```

```
'Salary': [30000, 40000, 55000, 65000, 90000, 110000, 130000]
}
df = pd.DataFrame(data)
X = df[['Years of Experience']]
y = df['Salary']
```

4. Split the dataset into training and testing sets.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

5. Train a Linear Regression model.

```
model = LinearRegression()
model.fit(X_train, y_train)
```

6. Predict the salary for an employee having 7 years of experience.

```
pred_7_years = model.predict([[7]])
print(f"Predicted Salary for 7 years of experience: {pred_7_years[0]:.2f}\n")
```

7. Display:

- **Slope (Coefficient)**

- **Intercept**

- **R² Score**

```
y_pred = model.predict(X_test)
print(f"Slope (Coefficient): {model.coef_[0]:.2f}")
print(f"Intercept: {model.intercept_:.2f}")
print(f"R2 Score: {r2_score(y_test, y_pred):.4f}\n")
```

8. Plot the regression line.

```
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, model.predict(X), color='red', label='Regression Line')
plt.title('Salary vs Years of Experience')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.legend()
plt.show()
```

